American Computer Science League

2022-2023 ● Contest 2: Short Problems Solutions ● Junior Division

1. Prefix/Infix/Postfix Notation

+ − * 2 3 / 4 2 ^ 2 3 = + − (* 2 3) (/ 4 2) (^ 2 3)
= + − (2 * 3) (4 / 2) (2 ^ 3)
= + (− 6 2) 8
= + (6 − 2) 8
= (+ 4 8)
= 4 + 8
= 12

C. 12

2. Prefix/Infix/Postfix Notation

$\frac{A+B}{C} - \frac{B*C}{A-C} = (A\ +\ B)\ /\ C\ -\ B\ *\ C\ /\ (A\ -\ C)$
$= (A\ B\ +)\ /\ C\ -\ (B\ C\ *)\ /\ (A\ C\ -)$
$= ((A\ B\ +)\ /\ C)\ -\ ((B\ C\ *)\ /\ (A\ C\ -))$
$= (A\ B\ +\ C\ /)\ -\ (B\ C\ *\ A\ C\ -\ /)$
$= A\ B\ +\ C\ /\ B\ C\ *\ A\ C\ -\ /\ -$

A. AB+C/BC*AC−/−

3. Bit-String Flicking

10100 OR 01100 AND 01011 = 10100 OR (01100 AND 01011)
= 10100 OR 01000
= 11100

B. 11100

4. Bit-String Flicking

(LCIRC-2 (RSHIFT-1 (NOT 01110))) = (LCIRC-2 (RSHIFT-1 10001))
= (LCIRC-2 01000)
= 00001

C. 00001

5. What Does This Program Do? (Looping)

This program calculates the compounded interest on $1000 at a yearly rate of 10% for 3 years.
1st year $1000 * .10 = $100
2nd year $1100 *.10 = $110
3rd year $1210 *.10 = $121
Total interest $1331-$1000 = $331

D. 331